

Carl "Isa" Perez

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- **Github** : evolvingsewage
- **Website**: perez.wiki

Education

- **2010-2014: C.E. Byrd High School**
- **Aug '14 – Feb '19 : Louisiana Tech University**
 - Major: Computer Engineering



Work Experience (see second page for additional details)

- **Feb '16 – Jun '19 : DevOps Consultant - Fenway Group**
 - Development of Continuous Delivery, Integration, Monitoring, & Failover systems for the IFS Maintenix application with Chef, Python, BASH, and Jenkins for Southwest Airlines. Configured Tomcat to host JasperReports for analytics in a HA environment.
 - Development of a Node.JS Web Application for Wellington Financial to communicate with MySQL, Oracle, and Postgres databases for internal financial data management.
- **Jul '19 - Present : Unix Systems/Linux Platform Engineer - CU Boulder**
 - Development and maintenance of continuous integration, continuous deployment, and monitoring systems using Ansible, Chef, Nagios/Nagios, Terraform, Github Actions, Openshift, and Jenkins for a broad set of teams across the University.
 - Leading upgrades of large swathes of obsolete RedHat systems with diverse requirements, from small faculty websites to the Dartmouth Flood Observatory's ArcGIS application.
 - 24/7 oncall support for critical services across the university, including its website, databases, and various student, faculty and research facing services.

Skills/Expertise

PROJECTS

Personal Website

A small demonstration of skills with Python, web development and deployment, and basic CD tools to introduce myself.

Perez Wiki

Senior Design

Python & Matlab programs designed by a small team to detect, process, & blend video & images.

Cyber Vision

Cloud Computing

A TensorFlow-based image processing cluster that splits the load of identifying a breed of dog from an image across multiple Pi's.

BCIC

METHODOLOGIES

Agile, Scrum, Kanban

ORGANIZATIONS

NAMI - President of the National Alliance on Mental Illness on Louisiana Tech's Campus – 2017-2018.

PROGRAMMING

Automation - Python, BASH, Ruby, YAML
Application Development - Java, Python
Web Development - NodeJS, React, Flask, REST, Bootstrap

TECHNOLOGIES

AI/LLM - Claude Code/Cowork, ChatGPT
Containers - Tomcat, WebLogic, Unicorn
CI + CD - Jenkins, Chef, Ansible, GitHub Actions
IDE - Vivado, Arduino, Visual Studio, Pulsar
OS - Arch Linux, Debian, CentOS, RedHat
Source Control - Git, SVN, BitBucket/Stash
Tracking & Ticketing - Service-Now, JIRA, MH
Web Standards - WSSE, JSON, SOAP

Detailed Work Experience

Feb '16 – Jun '19 : DevOps Consultant - Fenway Group

- Hired onto the MX DevOps team contracted to Southwest Airlines as a small part of the "largest single MRO system migration in the history of the airline industry." Initial focus was maintaining the Legacy Maxi-Merlin application while we created the develop and operations pipeline for the new application. The production environment was running on mainframe, and was becoming cumbersome to maintain. Additionally, SWA used TRAX to track another class of aircraft, and having two different application stacks to maintain was inefficient. Lastly, some aircraft maintenance processes were done on paper.
- The IFS Maintenix application was selected to replace all three separate maintenance tracking streams with a single high availability application. The DevOps team created the entire development stack, using Chef for configuration management and Jenkins for continuous integration. We set up testing pipelines for JUnit, Selenium, SoapUI, and JMeter. We were responsible for deployment of all parts of the new IFS Maintenix application across the Dev, Test, QA, and Production environments.
- My primary focus was supporting the Selenium testing against JasperReports, the deployment of JasperReports, SoapUI tests against various Maintenix Service Endpoints, and the maintenance of the CD/CI pipeline. Notably, I wrote a basic monitoring program in Python called Vostok to show the up/down status of all parts of the Maintenix software stack in all environments, served on a Flask Web Server.
- For a six month period in the latter half of 2018, I was concurrently contracted to Wellington Financial to help develop a NodeJS application to consolidate financial information spread across several different kinds of databases.

Jul '19 - Present : Unix Systems/Linux Platform Engineer - CU Boulder

- Hired onto the Unix Systems team for a 2 year appointment to maintain OIT's Linux systems. Team had a "wear many hats" approach, so I had many rotating responsibilities. Immediately assumed joint responsibility of the development of a new custom paging service that pulled data from Service-Now and used sendmail to page oncall users. System eventually migrated to postfix after OS upgrades. Supported several deployments of the colorado.edu website. Supported several RedHat upgrades, from versions 6 to 8, 7 to 9. Most of these upgrades were small faculty websites and applications, but also supported a few larger upgrades as well.
- Largest project was the Dartmouth Flood Observatory, a project at INSTAAR to acquire and preserve for public use a digital map record of the Earth's changing surface water, including changes related to floods, droughts, wetlands, shorelines, lakes, and reservoirs. I supported the upgrade of the ArcGIS application, MediaWiki documentation application, and various Tomcat containers from their versions on RedHat 6 to officially supported ones on RedHat 8.
- Eventually, role evolved into a Linux Platform Engineer. My focus narrowed from the entire campus to supporting OIT. I now primarily support OS upgrades, the maintenance of our CI/CD pipelines, and RedHat Openshift clusters. My current projects include shifting our TLS deployments to current best practices, continued maintenance of the pager system, and upgrades from RedHat 8 to 10.